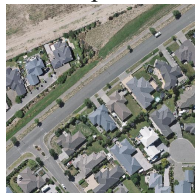


Input



Shared Feature & Proposal Extraction

SAM2 Hiera
BackbonePAFPN
(Neck)Multi-scale
FeaturesRPN &
ROI HeadDetected Box b_r

Mask ROI Feature

Output



Coarse Shape Prompt Mining(CSPM)

Mask ROI Feature

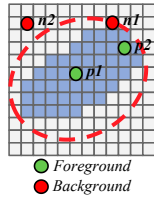
Coarse Mask
DecoderConv Block $\times 2$ Upsample $\times 2$

Conv Block

Upsample $\times 2$

Conv

Coarse Mask Logits

Detected Box b_r Adaptive 2P2N
Sampler

Resize

Paste

Sparse Prompts

Detected Box b_r

2P2N Points

Dense Prompts

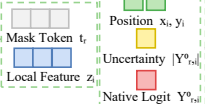
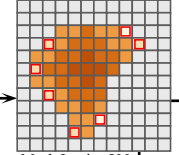
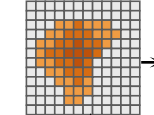
Uncertainty-Guided Decoder-Tail Refinement (UDR)

Image
Embeddings

SAM2 Prompt Encoder

SAM2 Mask Decoder

High Low

Refined Mask Logits Y_r^1 

Post-processing

